



BLUEWHALE STACK

FUNCTIONAL SPECIFICATION DOCUMENT

Corporate Website — FSD & UI/UX Design

Functional requirements, navigation architecture, and the complete UI/UX design system for the BlueWhale Stack website — aligned to the BlueWhale Stack Identity Guidelines v0.1101.

DOCUMENT	VERSION	DATE	COMPANION TO
Functional Spec (FSD)	1.0	14 June 2026	Website Scope of Work v1.0

About this document

Title	BlueWhale Stack — Website Functional Specification Document & UI/UX Design
Version	1.0
Date	14 June 2026
Relationship	Companion to the Website <i>Scope of Work</i> v1.0 — the SOW defines <i>what & why</i> ; this FSD defines <i>how it behaves and looks</i> .
Brand basis	BlueWhale Stack Identity Guidelines v0.1101 (2025) — logomark, logotype & primary blue applied throughout.
Audience	Design, front-end engineering, QA, and stakeholders approving the experience.

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Introduction, scope & conventions

1.1 Purpose

This Functional Specification defines, in implementable detail, the behaviour, structure, and visual design of the BlueWhale Stack corporate website. It is the contract between design and engineering: every page, navigation element, component, state, and interaction is specified so the build is unambiguous and testable.

1.2 Scope

In scope: the public marketing website — all pages, the global navigation (header, mega-menus, footer), the design system, responsive behaviour, forms, and non-functional standards. The Partner Portal application itself is a separate product; the website *links into* it and markets it, but does not re-implement it.

1.3 Conventions

- **FR-XXX** — a numbered functional requirement (testable).
- **Must / Should / May** — priority per RFC 2119 (Must = required for launch).
- Spacing values use an 8 px base scale; breakpoints are mobile-first.

1.4 Technology baseline

Next.js 15 (App Router) + TypeScript + Tailwind CSS v4, with a typed content layer and a CMS-ready seam (per the SOW). Server-rendered for SEO and speed.

Brand-aligned. All color, logo, and typographic direction in this FSD derive from the supplied BlueWhale Stack Identity Guidelines v0.1101. The horizontal logo, the whale-in-stacked-blocks logomark, and primary blue (#002da1) are the canonical identity; this document specifies how they are applied on the web.

SECTION 2

Users, roles & key journeys

The site is public (no authentication). "Roles" here are visitor intents the UX must serve. Authenticated experiences live in the product and Partner Portal, which the site links to.

Visitor type	Primary goal	Key journey
Evaluator (CIO / architect)	Assess fit, depth, credibility.	Home → Products → Edition → Module → Book demo.
Economic buyer (FinOps/CFO)	Quantify value & cost.	Solutions (FinOps) → Pricing → Contact.
Industry buyer (Gov/Telco/BFSI)	See sector & compliance fit.	Industry → relevant Edition → Demo.
Prospective partner	Understand programme & economics.	Partner → Portal showcase → Apply.
Talent	Evaluate company & roles.	About Us → Careers → Apply.
Existing customer/partner	Reach login / docs / support.	Utility nav → Login / Docs.

Every journey converges on one of three conversion endpoints: **Book a demo**, **Become a partner**, or **Contact sales**. These are the persistent CTAs the navigation and pages drive toward.

SECTION 3

Information architecture & sitemap

The site is organised under **five primary navigation buckets** plus a utility tier. The five buckets are exactly the items requested — **Products, Solutions, Industries, Partners, About Us** — each mapping to a clear body of content.

Nav bucket	What lives under it
Products	Platform overview, the 5 Editions (Standard, Enterprise, Datacenter, Government, Telco), the Module catalog (17 modules), adjacent products (WhaleForge, Pandava OS, Fincore), and Pricing.
Solutions	The 6 outcome solutions (Migration, FinOps, Security & Compliance, Automation, Data Center Management, Sovereign Cloud).
Industries	The 8 verticals (Government, Telco & MSP, BFSI, Healthcare, Manufacturing, Energy, Media, Enterprise SaaS).
Partners	Partner programmes, the Partner Portal showcase, "Become a partner" apply flow, and partner login.
About Us	Company & story, leadership, the product family, careers, newsroom, contact.
Utility (secondary)	Resources, Docs, Login, Region, Search, and the persistent "Book a demo" CTA.

3.1 Recommended primary order

Best-practice for a B2B platform leads with the offering and ends with the company. Recommended left→right order:

Products · Solutions · Industries · Partners · About Us

All five items you listed are present; only the order is optimised (Products first, About last). If you prefer your stated order it is a one-line change — see FR-NAV-002.

3.2 Full sitemap

Route	Page	Bucket
/	Home	—
/platform	Platform overview	Products
/editions + /editions/{5}	Editions overview + 5 detail	Products
/modules + /modules/[slug]	Module catalog + 17 detail	Products
/products/whaleforge pandava-os fincore	Adjacent product pages	Products
/pricing	Pricing & comparison	Products
/solutions + /solutions/[slug]	Solutions + 6 detail	Solutions
/industries + /industries/[slug]	Industries + 8 detail	Industries
/partners	Partner programmes	Partners
/partners/portal	Partner Portal showcase	Partners
/partners/apply	Become a partner (form)	Partners
/about	Company & story	About

Route	Page	Bucket
<code>/about/leadership</code>	Leadership & team	About
<code>/careers</code> + <code>detail/apply</code>	Careers	About
<code>/newsroom</code>	News & press	About
<code>/contact</code>	Demo & contact	About / global CTA
<code>/resources</code> + <code>/resources/[slug]</code>	Resource library	Utility
<code>/docs</code>	Documentation hub	Utility
<code>/legal/privacy</code> , <code>/legal/terms</code>	Legal	Footer

URL rules: lower-case, hyphenated, no trailing slash; collection at `/x`, detail at `/x/[slug]`; stable slugs (used in sitemap & analytics).

Navigation & header design

This section specifies the global header and how each of your five primary items is positioned and behaves.

4.1 Header anatomy

DESKTOP HEADER — 72 PX TALL, FULL-WIDTH, STICKY

BlueWhale Stack

(logo, left)

Products ▾

Solutions ▾

Industries ▾

Partners ▾

About ▾

Search

Login

Book a demo

Left: horizontal logo (links to Home). Centre/left-aligned: 5 primary items with caret = has mega-menu. Right: utility (Search, Login) + primary CTA button.

- FR-NAV-001** The header Must be sticky, reducing to 64 px and gaining a solid white background + subtle shadow after 8 px scroll.
- FR-NAV-002** Primary items Must render in the configured order (default: Products, Solutions, Industries, Partners, About). Order is data-driven (one array).
- FR-NAV-003** On dark hero pages the header May start transparent with the white logo, switching to solid on scroll.
- FR-NAV-004** The "Book a demo" CTA Must be persistently visible at all breakpoints.

4.2 Logo usage (per Identity Guidelines)

Header

Horizontal lockup, ~150 px wide.
Primary blue on light header;
white on transparent/dark.

Footer / mobile

Horizontal (white) in footer;
logomark-only as the mobile
tab/app icon & favicon.

Clear space

Maintain the guideline clear space
(x on all sides); never recolour,
stretch, or place on low-contrast
backgrounds.

4.3 Mega-menu specifications

Items with a caret open a **mega-menu** on hover (desktop) / tap, and via keyboard focus. Each opens a full-width panel under the header with columns + a promotional "feature" rail on the right.

Products — mega-menu

PLATFORM	EDITIONS	MODULES	FEATURED
Platform overview	Standard	Foundation	Compare editions → WhaleForge (IaC) Pandava OS Fincore Pricing
How it works	Enterprise	Operations	
Whale AI	Datacenter	Builder	
Security & trust	Government	Datacenter	
	Telco & MSP	Whale AI	

FR-NAV-010 Modules column links to grouped catalog; "Featured" rail promotes the edition comparator and adjacent products.

Solutions — mega-menu

BY OUTCOME		BY DEPLOYMENT	FEATURED
Cloud Migration Factory	Cloud Automation & DevOps	SaaS	FinOps: 20–30% in 90 days → View all solutions
FinOps & Cost Intelligence	Data Center Management	BYOC	
Security & Compliance	Sovereign Cloud	Sovereign / air-gapped	

REGULATED Government BFSI Healthcare	OPERATORS Telco & MSP Energy & Utilities Manufacturing	DIGITAL Media & Entertainment Enterprise SaaS	FEATURED Sovereign cloud for govt → View all industries
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Partners — mega-menu

PROGRAMMES Technology partners Channel & resellers Managed service providers	PARTNER PORTAL Portal overview Deal registration Margins & invoicing	GET STARTED Why partner Become a partner Partner login ↗	FEATURED Tiers: Silver / Gold / Platinum → Partner Portal tour
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About Us — dropdown (simple)

COMPANY Our story & mission Leadership & team The BlueWhale family	CONNECT Careers Newsroom Contact	FEATURED We're hiring → Certifications & trust
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4.4 Mega-menu behaviour

- **FR-NAV-011** Opens on hover after a 120 ms intent delay; closes on mouse-leave after 200 ms or on Esc.
- **FR-NAV-012** Fully keyboard operable: items are buttons with `aria-haspopup` / `aria-expanded`; arrow keys move within the panel; focus trap while open; Esc returns focus to the trigger.
- **FR-NAV-013** Only one mega-menu open at a time; opening another closes the previous.
- **FR-NAV-014** Panel animates with an 8 px fade-slide (150 ms); respects `prefers-reduced-motion`.

4.5 Utility navigation

A secondary tier (right of header, and a thin top bar on wide screens) carries items outside the five buckets: **Resources**, **Docs**, **Search**, **Login** (to product / Partner Portal), and optional **Region/Language**. This keeps the five primary buckets clean while these high-intent links stay reachable.

4.6 Mobile navigation (< 1024 px)

- **FR-NAV-020** A hamburger opens a full-height drawer; the five buckets render as **accordions** revealing the same links as the desktop mega-menus.
- **FR-NAV-021** Utility links (Resources, Docs, Login, Search) sit below the accordions; "Book a demo" is pinned to the drawer footer.
- **FR-NAV-022** Body scroll locks while the drawer is open; the drawer is keyboard- and screen-reader-accessible.

4.7 Footer

FOOTER — DARK (INK) BACKGROUND, 4 LINK COLUMNS + BRAND + NEWSLETTER			
Products Platform · Editions · Modules · Pricing	Solutions / Industries 6 solutions · 8 industries	Company About · Careers · Partners · Contact	

- **FR-FOOT-001** Footer Must contain: brand + tagline, newsletter signup, four link columns (Products, Solutions/Industries, Resources, Company), a compliance badge row, social links, and legal links.
- **FR-FOOT-002** Footer is identical across all pages and uses the white horizontal logo on the ink background.

SECTION 5

Functional requirements (per page)

Each page's behaviour as testable requirements. Layout/visual detail is in §6–9; this section is the functional contract.

5.1 Home /

ID	Requirement
FR-HOME-001	Hero Must present the value proposition, dual CTA (Book a demo / Explore platform), and a trust strip (clouds, regions, compliance).
FR-HOME-002	Must render sections in order: hero, cloud strip, problem→promise, capabilities (6), editions teaser (5), industries strip (8), how-it-works (4), outcomes stats, testimonials (3), resources teaser, closing CTA.
FR-HOME-003	Stat counters Should animate on scroll into view (reduced-motion: show final value).
FR-HOME-004	Every card linking to a sub-page Must be a single accessible link with a visible focus state.

5.2 Products — Platform, Editions, Modules

ID	Requirement
FR-PROD-001	Platform page Must include an interactive module map (control plane over clouds), deployment models, and Whale AI tiers.
FR-EDN-001	Editions overview Must render the 5 editions and a module×edition comparison matrix sourced from content data.
FR-EDN-002	An edition comparator Must let users select/compare editions and filter the matrix.
FR-EDN-003	Each edition detail page Must show hero, positioning, included-modules checklist, architecture diagram, key specs (deploy/AI tier/price), and CTA.
FR-MOD-001	Modules overview Must group modules (Foundation/Operations/Builder/Datacenter/AI); each detail page shows description, features, "available in" editions, and a diagram.
FR-MOD-002	Unknown edition/module/industry/solution slugs Must return the 404 page.

5.3 Solutions & Industries

ID	Requirement
FR-SOL-001	Each solution detail Must present problem, approach, architecture diagram, measurable outcomes, who-it's-for, and related modules.
FR-IND-001	Each industry detail Must present challenges, why-BlueWhale, KPIs, compliance mapping, targets, and a relevant-edition CTA.

5.4 Partners & Partner Portal showcase

ID	Requirement
FR-PRT-001	Partners page Must present the 3 programmes (Technology, Channel/Reseller with tier margins, MSP) and "why partner" benefits.
FR-PRT-002	Partner Portal showcase Must walk through deal registration → two-sided invoicing → provisioning, with annotated visuals

ID	Requirement
FR-PRT-003	"Become a partner" Must open an application form; "Partner login" Must deep-link to the Partner Portal (new tab).

5.5 About, Careers & Contact

ID	Requirement
FR-ABT-001	About Must present story/mission, principles, the product family, by-the-numbers, global offices, leadership, and certifications.
FR-CAR-001	Careers Must list open roles (filter by department/location/remote) and provide an application form with résumé link.
FR-CON-001	Contact Must provide a demo-request form, contact channels, and offices; submission triggers FR-FORM-.*.

5.6 Resources, Docs, Pricing, Legal

ID	Requirement
FR-RES-001	Resources Must be filterable by type (whitepaper, case study, webinar, blog, video, guide); detail pages May gate downloads behind a short form.
FR-PRC-001	Pricing Must show edition tiers + comparison + FAQ; figures controlled by the "publish vs contact-sales" setting.
FR-DOC-001	Docs hub Must link out to product/API documentation.

5.7 Global & system pages

ID	Requirement
FR-GLB-001	Every page Must include header, footer, correct metadata/OG/JSON-LD, and exactly one primary CTA path.
FR-GLB-002	A persistent assistant widget Should help route visitors (editions, FinOps, demo).
FR-SYS-001	404 and 500 pages Must be branded and offer Home + Contact links; <code>sitemap.xml</code> & <code>robots.txt</code> Must be generated.

UI/UX design system — brand foundations

The visual language extends the BlueWhale Stack Identity Guidelines v0.1101 to the web: the same primary blue, the architectural logic of the stacked-blocks logomark, and bold, geometric typography — executed as a refined, modern-enterprise system.

6.1 Colour

Primary blue `#002da1` is canonical (from the identity). The web palette adds a tonal blue scale, a cyan accent for highlights/interaction, a neutral slate scale, and semantic colours.

Role	Token	Hex	Sample
Brand / primary	brand-600	#002da1	
Brand deep (hero/footer)	brand-900	#00154b	
Brand hover	brand-700	#00237f	
Accent (interaction)	accent-500	#06b6c4	
Ink (dark surface)	ink-950	#060a18	
Text	slate-900	#0f172a	
Muted text	slate-500	#64748b	
Surface / line	slate-50 / 200	#f8fafc / #e2e8f0	

Open brand item. The supplied guidelines excerpt covers the logo only. The *official* secondary palette and exact accent should be confirmed from the full brand guide; the cyan accent above is a recommendation and is trivially swappable to the brand's defined secondary.

6.2 Typography

The wordmark uses a **bold, squared, geometric sans-serif** (per the identity). The web type system mirrors that character for display while keeping body text highly legible.

Use	Typeface	Notes
Logo / wordmark	Brand wordmark (supplied)	Used only as the logo asset, never re-typeset.
Display / headings	Geometric techno sans — recommend Chakra Petch or Space Grotesk (on the licensed brand font)	Echoes the squared logotype; used for H1–H3 & stats. BlueWhale Stack · Website (optimal) · Confidential · v1.0

Use	Typeface	Notes
Body / UI	Inter	Neutral, legible at all sizes; weights 400/500/600/700.
Mono (code/specs)	JetBrains Mono / system mono	Code snippets, API references.

FR-TYP-001 Type scale (fluid, `clamp()`): H1 36–60px · H2 28–40 · H3 20–26 · body 16–18 · small 13–14. Line-height 1.1–1.25 headings, 1.55 body. Max line length ~70ch.

Open brand item. Confirm the licensed display/brand font from the full guidelines; the recommendations above are close geometric matches and a one-line swap.

6.3 Grid, spacing & radius

Grid

12-column, 1280px max content width, 24px gutters; 20px outer padding (mobile), generous section rhythm (80–96px desktop).

Spacing

8px base scale (4/8/12/16/24/32/48/64/96). Consistent vertical rhythm between sections.

Radius & elevation

Cards 16px, buttons 8px, pills 999px. Two shadows: soft (cards), brand-glow (primary/hover). Echoes the rounded stacked-blocks.

6.4 Logomark as a design motif

The stacked-blocks-with-whale logomark is more than a logo: its **2×2 block grid** becomes a recurring visual device — section dividers, the "platform layers" diagram language, bento-style card grids, and loading states — tying the site visually to the identity.

Component library & layout patterns

7.1 Core components

Component	States / variants
Button	Primary, secondary, ghost, white; sizes sm/md/lg; default, hover (lift), focus-ring, active, disabled, loading.
Card	Static & interactive (hover lift + brand shadow); used for editions, modules, industries, resources.
Badge / Pill	Brand, accent, neutral tones; for tiers, tags, "new".
Stat	Value + label; light & inverse; animated counter.
Tabs / Accordion	For module groups, FAQ, mobile nav; keyboard accessible.
Comparison table	Sticky header row, ✓/– cells, responsive horizontal scroll.
Mega-menu	Columns + featured rail (per §4.3).
Form controls	Input, select, textarea, checkbox, consent; label, helper, error states.
CTA band	Gradient brand section with dual CTA (reused site-wide).
Diagram / Placeholder	Code-based SVG architecture diagrams; branded placeholder until assets land.

7.2 Page layout patterns



- **Hero patterns:** dark "ink" hero for Home/edition/solution; light hero for overview/industry pages.
- **Section rhythm:** alternating white / slate-50 backgrounds to separate sections; one idea per section.
- **Bento grids:** capability & module overviews use bento-style cards echoing the block motif.

SECTION 8

Interaction & motion design

Pattern	Spec
Entrance	Fade + 14px rise on scroll-into-view; 500–600ms; staggered ~60ms across a group.
Hover (cards/buttons)	2–4px lift + shadow; 150–200ms ease-out.
Mega-menu	8px fade-slide, 150ms (per FR-NAV-014).
Stat counters	Count up over ~1.2s when in view.
Interactive diagrams	Hover/tap reveals node detail; smooth path transitions.
Reduced motion	All of the above degrade to instant/no-motion under <code>prefers-reduced-motion</code> .

FR-MOT-001 Motion Must be purposeful and subtle — guiding attention, never blocking content or harming performance. No autoplaying carousels.

Responsive & breakpoint behaviour

Breakpoint	Width	Key behaviour
Mobile	< 640px	Single column; hamburger drawer nav; stacked cards; full-width CTAs.
Tablet	640–1023px	2-column grids; still hamburger nav; comparison tables scroll horizontally.
Desktop	1024–1279px	Full mega-menu header; 3-column grids.
Wide	≥ 1280px	Max 1280px content; optional top utility bar; 4–5 column grids.

FR-RES-010 All interactive targets Must be ≥44×44px on touch. Content parity across breakpoints — nothing hidden on mobile that exists on desktop (except decorative).

Forms, validation & messaging

ID	Requirement
FR-FORM-001	Demo/contact form fields: first & last name, work email, company, phone (opt), job title (opt), industry, edition interest, message, consent checkbox.
FR-FORM-002	Inline validation on blur + on submit; clear, specific error messages; email format & required-field checks; consent required.
FR-FORM-003	Submit posts to a Next.js route handler; success shows a confirmation with reference; failure shows a retryable error. Notification email sent to sales.
FR-FORM-004	Spam protection (honeypot + rate limit). No secrets client-side.
FR-FORM-005	Careers & partner-apply forms reuse the same control library & validation patterns.

Non-functional requirements

Area	Requirement
Performance	Lighthouse ≥ 90 all categories; LCP $< 2.5s$, CLS < 0.1 , INP $< 200ms$ on mobile 4G; images optimised; fonts <code>display: swap</code> .
Accessibility	WCAG 2.1 AA: semantic landmarks, alt text, keyboard operability, visible focus, AA contrast, reduced-motion, ARIA on nav/menus.
SEO	SSR/SSG, per-page title/description/OG, JSON-LD (Organization/Product/BreadcrumbList), canonical, sitemap & robots.
Analytics	Privacy-respecting analytics; conversion events on every CTA & form submit.
Security	Security headers/CSP, HTTPS, form spam protection, dependency hygiene.
Browser support	Latest 2 versions of Chrome, Edge, Safari, Firefox; iOS/Android.
Maintainability	Typed content layer + CMS-ready seam; documented component library.

Acceptance criteria & open brand items

Acceptance (launch gate)

- All FR-* "Must" requirements implemented and verified.
- Navigation (header mega-menus, mobile drawer, footer) works per §4, keyboard + screen-reader tested.
- Design system applied consistently; brand logo & colour per identity guidelines.
- Lighthouse ≥ 90 , WCAG AA, cross-browser sign-off; forms submit & notify.

Open brand items (need from you)

1. **Full brand guide pages** beyond the logo — the official *secondary colour palette* and *typography system* (so we replace the recommended accent & display font with the canonical ones).
2. **Licensed brand/display font files** (or confirmation to use the recommended Chakra Petch / Space Grotesk).
3. **Final primary nav order** — confirm recommended (Products-first) or your stated order.
4. **Imagery, leadership content, customer logos & KPI confirmation** (carried from the SOW).

End of FSD v1.0. This specification, with the Scope of Work v1.0, is sufficient to begin building the world-class site. The logo and primary blue from your identity guidelines are already integrated into the project (`public/brand/`).